

Vulnerability Assessment Report

22nd May 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 2026 to August 2026. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The database server contains important business data; data of the organization, private data of its employees and their customers private data. It is very important to secure the server data in order to preserve the reputation and finances of the company, comply with the law and regulations and keep their employees and clients safe.

Another thing that we should look for is to make the server robust, so that it can provide database management services without interruptions.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Competitor	Conduct Man-in-the-middle attack (MITM)	1	2	2
Employee	Alter/Delete critical information	1	3	3
Competitor	Conduct Denial of Service (DoS) attack	2	3	6

Approach

Conducting Man-in-the-middle (MITM) attack would most likely be tried by a competitor, but the likelihood of the threat event happening is highly unlikely due to the fact that the server uses the newest Linux OS distribution that is completely patched. SSL/TLS is also another factor which influences the likelihood and severity to be low. Competitors could also try to conduct Denial of Service (DoS) attacks, which would be a more severe event due to the importance of database management server being constantly online. The likelihood of this event happening is regarded as medium due to the good security policies implemented by the organization in combination with powerful hardware configuration.

At last, employees could accidentally or intentionally alter or delete critical information. The likelihood of this scenario happening is low, due to the education of employees and proper access management. However, the severity of the incident is high, due to the importance of data.

Remediation Strategy

It is very important to educate the personnel, establish least privilege, password policies, defense in depth, and conduct security audits regularly. Due to the high importance of the data and service provision it is also necessary to have regular backups and fast restore possibilities.